

Who Hauls the Habitat? A Simulation Testbed and a Pooled Dispatch Marketplace for Lunar Surface Logistics

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Roatify Research · Preprint, October 2026 · Code, scenario files, seeds and figure scripts: [lunar-logistics-research/](https://github.com/roatify/research/)

Abstract. NASA's Moon Base plan lands about 4 tonnes of cargo at the lunar south pole before 2029, roughly 60 tonnes in 2029–2032 and 150 tonnes after that. Individual items range from 10 kg instruments to 15-tonne habitats, yet the rovers under contract carry 800–1,600 kg, and NASA's own architecture papers say that no planned vehicle can relocate a large element. NASA's April 2026 architecture guide names “a marketplace for lunar logistics” as a goal but does not say what one would do. We study that question with a discrete-event testbed of surface cargo movement at a nine-site south-pole base. The testbed models lander arrivals and manifests, a heterogeneous fleet owned by several companies, the lunar night, Earth–Moon control latency, vehicle faults, vehicles' other duties, and the return flow of trash and samples. We compare four dispatch rules: bilateral in-house fleets (today's contracting), pooled greedy dispatch, batched optimal assignment, and a pooled marketplace built on a sequential single-item auction with bundling, consolidation of bulk lots, team lifts for oversize items, backhaul and night-aware acceptance. Across 30 common-random-number replications per cell we find three things. First, pooling matters only when the market is fragmented: with one owner the marketplace and in-house dispatch are statistically indistinguishable, but with six owners sharing six vehicles the in-house 90th-percentile delay of urgent cargo is 40 h longer than the marketplace's. Second, the marketplace reaches a 90% same-day service level for urgent cargo with four vehicles at Phase-2 demand where greedy pooled dispatch needs six and in-house fleets do not reach it with 8; because the fleet is 99% of the cost, a surface move costs about \$300 per kilogram of mass moved, or about \$5,000 per kilogram of landed cargo, against a landed cost of \$78,000–\$1,000,000 per kilogram. Third, the heavy-cargo gap is real and cannot be closed by coordination alone: with LTV-class rovers only, 54% of landed mass is immovable, team lifts recover under half of it, and one 10-tonne hauler removes most of the rest. Teleoperation from Earth is workable at lunar latency but collapses at Mars-like delays, where delivered mass falls below 10%. We release the testbed, scenarios, seeds and all scripts.

Keywords: lunar surface logistics, multi-robot task allocation, auction mechanisms, discrete-event simulation, horizontal collaboration, Artemis, Moon Base

1. Introduction

A decade of lunar planning has settled where cargo will land and how much of it there will be. NASA's *Moon Base Architecture User's Guide* (April 2026) describes three phases: 21 landings delivering about 4,000 kg to the south pole in the first, 24 landings and about 60,000 kg in the second, 28 landings and about 150,000 kg in the third [1]. The agency's 2024 architecture white papers put recurring logistics at 2,000–6,000 kg per crewed mission, and occasional single elements such as habitats and pressurised rovers at 12,000–15,000 kg [2, 3]. Landing pads must sit more than a kilometre from the habitation zone because of plume ejecta, so everything that lands has to be moved, some of it as far as 5 km over slopes of up to 20 degrees [2].

Far less has been settled about who moves it. The two Lunar Terrain Vehicles (LTVs) ordered in May 2026, from Astrolab and Lunar Outpost at about \$220 million each, are crew rovers that carry 800 kg of cargo at full performance and roughly 1,600 kg at reduced performance [4, 5]. NASA's 2025 logistics workshop states the mismatch plainly: “no mobility assets exist that can relocate large elements” [5]. Each vehicle will belong to one company under one contract, and NASA has told the LTV providers they may sell rover time to other customers between NASA missions [6]. In January 2025 NASA paid nine companies \$24 million to study lunar logistics, including “surface cargo and mobility” and “business models” [7]. And on page 14 of the 2026 guide, the agency writes that “fostering a marketplace for lunar logistics will empower government and industry to develop the capabilities and operational competencies needed to support crewed missions to the Red Planet” [1].

That sentence is a goal, not a design. Nobody has said what a surface logistics marketplace would dispatch, with what rule, under which constraints, or whether it would help at all when the whole market is two rovers and a dozen cargo items a year. The question is not academic for us: Roatify is building a freight-dispatch platform for Earth trucking and intends to carry that software to the Moon in stages. Before raising money against that plan we wanted an honest answer to four questions.

1. **RQ1.** When does pooled dispatch beat bilateral in-house fleets, and by how much, for fleets of two to eight vehicles?
2. **RQ2.** How large is the heavy-cargo gap in delivered mass, and how much of it can coordination (several rovers lifting one item together) recover without a new vehicle class?
3. **RQ3.** How do autonomy level and control latency bound surface throughput, and how many Earth operators does a fleet need?
4. **RQ4.** What does a surface move cost per kilogram, how does it compare to the cost of landing that kilogram, and which assumptions drive it?

Earth freight gives reasons for both optimism and caution. Pooling fleets across shippers saves 5–30% of transport cost in road and forest logistics [8, 9, 10]. But those gains come from thick markets. Theory on dynamic matching says that in thin markets greedy matching is close to optimal and waiting to batch requests mostly adds delay [11, 12]; empirical work on US trucking ties the efficiency of the spot market directly to its thickness [13]. The lunar market will be the thinnest freight market anyone has ever tried to run.

We answer the four questions with simulation, because there is no data to answer them otherwise. We built a discrete-event testbed of cargo movement at a nine-site south-pole base and ran about 7,000 simulated years of operations. The testbed reproduces NASA's published spacing, slopes, cargo masses, lander cadence and rover capacities; it adds the lunar night, Earth–Moon control latency fitted to NASA's own teleoperation trials, vehicle faults, vehicles' other duties and a return flow of trash and samples. On top of it we implemented four dispatch rules, from today's bilateral contracting to a pooled marketplace that combines a sequential single-item auction with bundling, consolidation, team lifts, backhaul and night-aware acceptance. Our contributions are:

- We introduce **LunarSim**, an open discrete-event testbed for lunar surface logistics with three demand phases, three vehicle classes, a sourced parameter table and ten experiment suites. Each parameter carries its source and a confidence grade.
- We propose a **pooled dispatch marketplace** for multi-owner rover fleets and give its clearing rule as a generalised-cost sequential auction (Algorithm 1), with team lift for items heavier than any single vehicle.

- We show, with 30 common-random-number replications per cell and paired confidence intervals, **where pooling helps and where it does not**: the gain is zero for a single owner, grows with fragmentation and with the share of time vehicles spend on other duties, and reduces the fleet needed for a same-day service level from six to four vehicles at Phase-2 demand.
- We quantify **three limits that no dispatch rule removes**: immovable heavy cargo, the collapse of teleoperation at Mars-like latency, and the fleet's fixed cost, which sets the cost of a surface move regardless of how cleverly the fleet is dispatched.

Section 2 reviews what is known. Section 3 describes the testbed and its data. Section 4 defines the mechanisms. Section 5 gives the experimental protocol. Section 6 reports results, Section 7 discusses what they mean for a marketplace operator and for NASA, and Sections 8–10 cover limitations, impact and conclusions. Appendices give the full parameter list, the simulation reporting checklist and additional tables.

2. Background and related work

2.1 What NASA has published about surface cargo

Three NASA documents anchor our demand model. The *Lunar Mobility Drivers and Needs* white paper (2024) gives the range envelope (5–5,000 m), the ejecta standoff ($>1,000$ m), the slope envelope (up to 20° , with slopes over 10° “common”), the cargo classes (500–2,000 kg demonstrations, 2,000–6,000 kg recurring logistics per crewed mission, 12,000–15,000 kg habitation elements) and the capability expressed in the architecture (800 kg uncrewed on the LTV, about 1,500 kg planned in total) [2]. It lists the gaps the agency wants industry to fill: autonomy or tele-robotic operation through the whole year, standard cargo interfaces, offloading, levelling, mated power, and “the ability of multiple robotic mobility systems to work together” [2]. The *Lunar Surface Cargo* paper forecasts aggregate demand of 2,000–10,000 kg per year during the foundational phase, with occasional deliveries of up to 15,000 kg [3]. The 2026 architecture guide supplies the phase totals, a 10 km/h target for large rovers, a 2-tonne lander target for Phase 1, and the marketplace sentence quoted above [1].

2.2 Space logistics modelling

Campaign-level space logistics has a mature modelling literature. MIT's SpaceNet and the time-expanded network models of Ho, de Weck and colleagues optimise commodity flows between Earth, orbit and the lunar surface over mission campaigns [14, 15, 16]. These models decide how much mass to send and when; they treat the surface as a single node. The surface itself has been studied for traverse planning and rover energy budgeting (VIPER's planning deck gives speeds of 10–20 cm/s while driving and a mission average of 1 cm/s [17]), and for cooperative transport of large payloads at the control level [18, 19]. We found no model of dispatching cargo among several companies' vehicles on the surface, and no treatment of lander-arrival lumpiness, lighting windows and latency together.

2.3 Multi-robot task allocation and market-based coordination

Our mechanism sits in the market-based multi-robot task allocation (MRTA) tradition surveyed by Dias et al. [20] and formalised by Gerkey and Mataric [21]. The sequential single-item (SSI) auction of Koenig, Lagoudakis and colleagues allocates tasks one at a time to the robot whose route cost increases least; its total path cost is within a factor of two of optimal under symmetric, triangle-inequality costs, and it needs at most $|\text{tasks}| \times |\text{robots}|$ bids [22, 23]. The consensus-based bundle algorithm (CBBA)

decentralises the same idea [24]. These methods assume one cooperative owner and have no notion of price or of items that need two robots. We keep the SSI structure, replace route cost with a generalised cost that includes the shipper's value of time, and add a team-lift stage.

2.4 Freight marketplaces, collaboration and market thickness

Combinatorial auctions for truckload procurement [25], horizontal collaboration among carriers [8, 9] and digital freight matching [26] all report that pooling demand across shippers cuts cost, typically by 5–30%, and that the hard part is a fair split of the gain [9, 10]. Dynamic matching theory adds the thin-market caveat: without information about which requests are about to expire, greedy matching is near-optimal and batching mostly adds waiting [11]; a windowed matching that waits $d+1$ periods is only 0.279-competitive [12]. On US spot trucking, Harris and Nguyen find efficiency rising with thickness [13]. Even in that market only about 15% of loads clear on the spot exchange; the rest move under contracts [27]. We use these results to choose baselines (greedy and batched) and to design the thickness experiment.

2.5 Teleoperation under delay and supervisory control

Round-trip light time to the Moon is about 2.6 s; with relays and ground processing NASA quotes up to 10 s one way [28]. NASA JSC's LTV teleoperation trials measured average driving speeds of 3.24, 2.56, 2.03 and 1.76 km/h at 0, 4, 6 and 8 s of round-trip delay [29]. A lunar-excavation teleoperation study found task completion times of 139% and 174% of baseline at 1.5 s and 3 s delay, with success rates falling from 91% to 45% [30]. Operators adopt a move-and-wait strategy once delay exceeds about a second [31]. For supervisory control, Olsen and Wood's fan-out metric gives 1–9 robots per operator depending on autonomy [32], and mining autonomous haulage runs up to about 30 trucks per controller [33]. We fit the speed model in Section 3.4 to the JSC numbers and sweep operator fan-out in Section 6.5.

Table 1. Where this work sits relative to the nearest prior families. A tick means the family models the feature; a dash means it does not.

Family	Multi-owner fleet	Prices / bids	Oversize items (team lift)	Lunar night	Control latency	Lander lumpiness	Open code
Campaign logistics (SpaceNet, time-expanded networks) [14–16]	–	–	–	partly	–	✓	partly
MRTA auctions (SSI, CBBA) [22–24]	–	bids, no payments	–	–	–	–	partly
Freight matching and collaboration [8, 9, 25, 26]	✓	✓	–	–	–	–	–
Teleoperation studies [29–32]	–	–	–	–	✓	–	–
This paper (LunarSim)	✓	✓	✓	✓	✓	✓	✓

3. The testbed

LunarSim is a discrete-event simulation written in Python 3.12 with NumPy, SciPy and NetworkX. Time is measured in hours; a run covers one Earth year of operations. This section is a summary ODD

description [34]; the full parameter list with sources is in Appendix A and the STRESS reporting checklist [35] in Appendix B.

3.1 Purpose and entities

The model exists to compare dispatch rules for a shared pool of rovers that move landed cargo at a south-pole base, under the constraints NASA has published. Three kinds of entity flow through it. *Cargo items* have a mass, an origin pad, a destination site, a release time (when offloading finishes), a priority class (1 urgent, 2 normal, 3 deferrable), an owner, an interface standard and a class label. *Vehicles* have an owner, a class (capacity, speed, energy, battery, night capability, team-lift capability, unit cost, design life, fault parameters, handling times), a location, an autonomy level and a set of accepted interface standards. *Jobs* are what a dispatch rule returns: one or more vehicles, a bundle of items sharing an origin, a visiting order of destinations, and optional backhaul items. Figure 2 shows how the pieces connect.

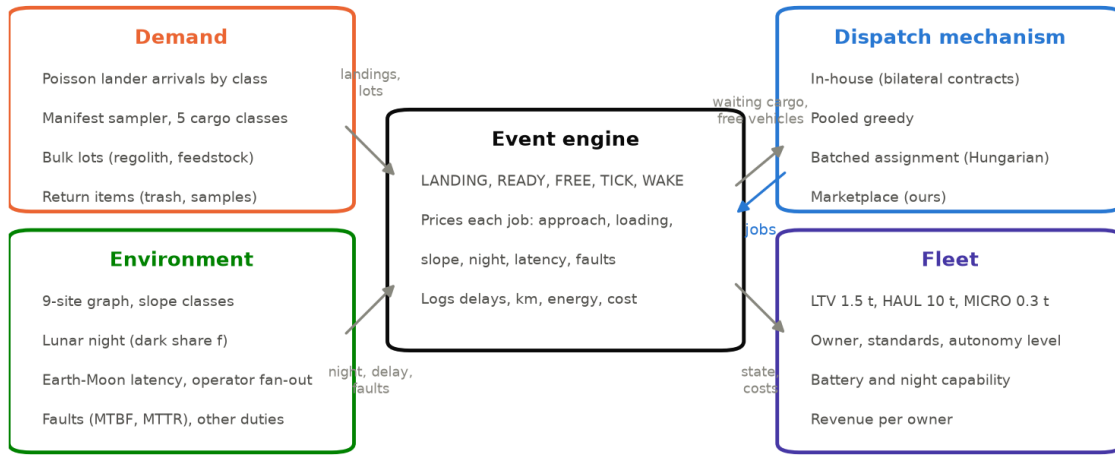


Figure 2. Testbed architecture. The demand and environment submodels feed an event engine that calls a dispatch mechanism whenever cargo becomes ready or a vehicle becomes free (or at fixed ticks for batched mechanisms). The engine prices every returned job honestly: empty approach, loading, slope- and night-adjusted travel, teleoperation slowdown, operator availability, random faults and a hibernation penalty if a vehicle without night capability is caught in the dark.

3.2 Sites and terrain

The base is a nine-site graph on a local tangent plane (Figure 1). Two landing pads sit 1.7 and 2.1 km from a cargo depot beside the habitation zone, satisfying the >1 km ejecta standoff. A power site, a construction and excavation site and an in-situ resource (ISRU) pilot plant lie 0.6–2 km from the base; a science site and a shadowed crater rim (PSR) lie about 3 and 5 km out, matching NASA's 5,000 m aggregation range. Each edge carries a slope class (flat, 0–5°; moderate, 5–10°; steep, 10–20°) that multiplies travel time by 1.0, 1.33 or 2.0 and energy by 1.0, 1.35 or 1.9. Straight-line distances are scaled by a tortuosity factor of 1.3 to represent paths that wind around craters and boulders. Vehicles take shortest-time paths.

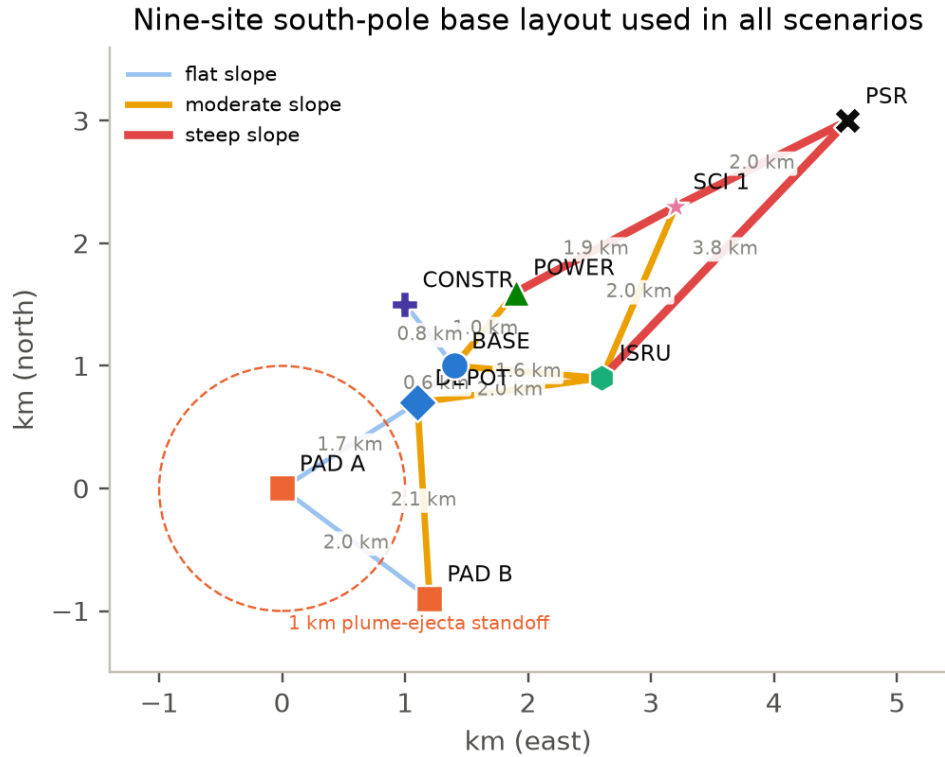


Figure 1. The nine-site south-pole base used in every scenario. Edge labels give path lengths after the tortuosity factor; colours give slope class. The dashed circle marks the 1 km plume-ejecta standoff that keeps pads away from the base.

3.3 Demand

Landers arrive as independent Poisson processes per class. CLPS-class landers deliver 100–500 kg (Firefly Blue Ghost, Intuitive Machines Nova-C, Astrobotic Griffin); MK1-class landers deliver 2,000–3,000 kg (Blue Moon MK1); large landers deliver 12,000–30,000 kg (Starship HLS cargo, Blue Moon MK2). Each landing draws a manifest from five cargo classes (Table 2) until the payload mass is used up, with the lander's crane limiting the heaviest single item. Items become available 6–72 h after touchdown depending on lander class, staggered by 15 minutes per item. Three phases scale the cadence to match NASA's totals: Phase 1 (about 3 t/yr of landed cargo, 14 items), Phase 2 (about 17 t/yr, 51 items) and Phase 3 (about 75 t/yr, 87 items). Figure 3 shows the manifest-based landed-mass forecast we assembled from public mission plans and the simulated item-mass distribution by phase.

Two further flows matter for a dispatch rule and are usually omitted. *Bulk lots* represent regolith moved for berms and shielding and excavated feedstock carried to the ISRU plant; they are released evenly through the year in 500 or 1,000 kg lots as deferrable (priority 3) items. NASA sizes its IPEX excavator demonstration at 10,000 kg of regolith per day [36], so our Phase-2 default of 500 t/yr and Phase-3 default of 1,500 t/yr are deliberately modest; Section 6.7 sweeps 0–3,000 t/yr. *Return items* represent trash, samples and empty carriers: 30% of outbound items generate a return item of 20–60% of their mass, ready 5–30 days after delivery, bound for a pad. Both are drawn from the demand random stream so that every mechanism sees the same flows.

Table 2. Cargo classes used by the manifest sampler. Masses are uniform within the range; destinations are drawn with the listed weights. Shares are per item for a build-out manifest; the lander's payload and crane limit truncate them.

Class	Mass (kg)	Destinations (weight)	Priority	Item share	Anchor
Consumables and spares	60–400	BASE 0.8, DEPOT 0.2	1	0.34	Logistics carriers; packaging overhead 75–150% [37]
Science payloads	10–150	SCI 0.5, PSR 0.3, BASE 0.2	2	0.20	CLPS payload masses 1–100 kg [38]
Equipment and demos	300–2,000	CONSTR 0.4, ISRU 0.3, POWER 0.2, BASE 0.1	2	0.26	“500-to-2,000 kg” demonstrations [2]
Infrastructure	2,000–6,000	POWER 0.4, ISRU 0.3, CONSTR 0.3	2	0.14	Logistics 2,000–6,000 kg per mission [2]
Elements	6,000–15,000	BASE 0.6, POWER 0.4	3	0.06	Habitats, power units 12,000–15,000 kg [2, 3]
Bulk lots (regolith, feedstock)	500 or 1,000	CONSTR to BASE / ISRU; PSR to ISRU	3	scenario	IPEx 10 t/day [36]
Return items	20–60% of parent	back to the origin pad	3	30% of items	Trash management, sample return [1, 7]

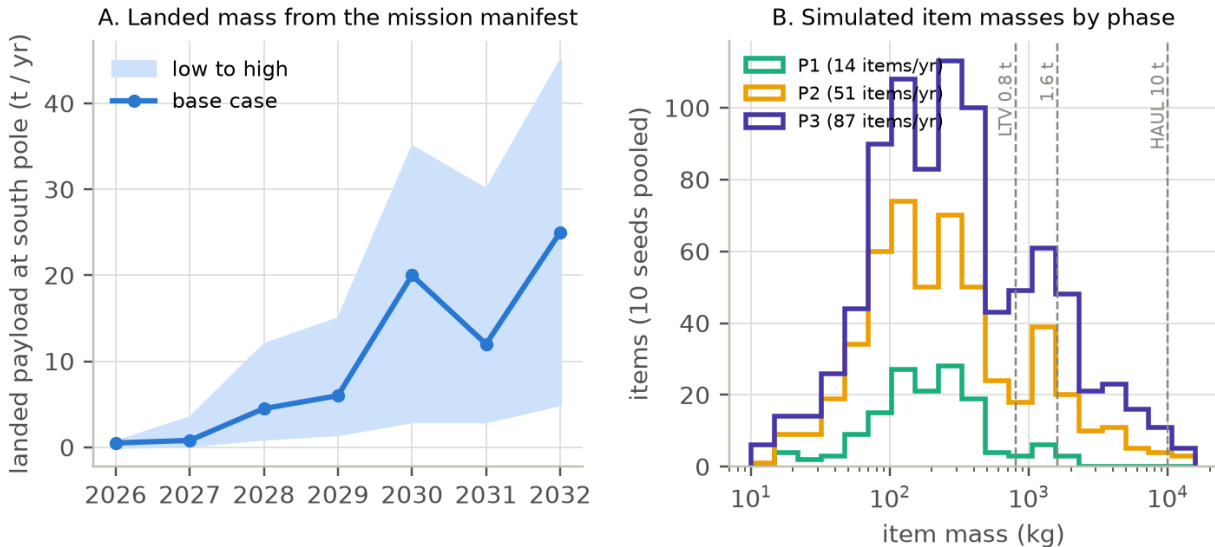


Figure 3. Demand. **A:** landed payload mass at the south pole by year, assembled from public mission manifests (low, base and high cases; Appendix A lists each mission). The 2026–2028 base case is five or six discrete sub-3-tonne items, which is the single most important fact for a marketplace design. **B:** item-mass histograms from ten simulated years per phase. Vertical lines mark the LTV’s 800 kg full-performance and 1,600 kg reduced-performance cargo limits and the 10-tonne hauler we introduce.

3.4 Environment submodels

Lunar night. A synodic month is 708.7 h. A fraction f of each cycle is dark at the base; the default is 0.15 (about 106 h), consistent with 90–94% illumination on the Shackleton rim [39] and NASA’s 120-hour darkness survival target [1]. Vehicles without night capability cannot start jobs in the dark; if one is caught out when night begins it hibernates until dawn, and the job is delayed by the whole night. Night-capable vehicles drive at 60% speed in the dark.

Control latency and autonomy. Each vehicle is teleoperated, supervised or autonomous. Round-trip control latency is 2.6 s of light time plus a ground-processing term, 2.4 s by default (RTT 5 s). We fit

the JSC trial speeds [29] with $v(\text{RTT}) = v_0 / (1 + k \cdot \text{RTT})$, which gives $k = 0.105 \text{ s}^{-1}$; the zero-delay teleoperated speed is 3.24 km/h, 0.54 of the 6 km/h autonomous average we assume for a 10 km/h-class rover. Teleoperated loading and unloading take $(1 + 0.25 \cdot \text{RTT})$ times nominal, fitted to the excavation study [30]. Supervised vehicles drive at 0.9 of autonomous speed with a 1.1 handling factor. Earth operators are a shared resource: a teleoperated vehicle needs one operator for the whole job, a supervised vehicle 1/4 of one (fan-out 4) and an autonomous vehicle 1/20. A job that cannot get its operator waits.

Faults and other duties. Each vehicle fails after an exponentially distributed number of operating hours (mean MTBF 300–500 h by class) and recovers after a fixed MTTR of 12–36 h; this yields 92–96% availability, in the range reported for autonomous mining haulage [40]. Vehicles also spend a share of their time (default 30%) on their owner's other work in 48-hour blocks, reflecting NASA's statement that LTVs serve NASA missions first and may be hired out in between [6].

Energy. Driving energy is 350 Wh/km for an LTV-class rover (an Apollo LRV-derived design value is about 95 Wh/km for a 700 kg vehicle [41]; we scale up for mass and slopes), rising with payload and slope class. A vehicle whose battery cannot cover a job charges first at 5 kW. Energy never binds in the scenarios reported here, so we do not discuss it further.

3.5 Vehicles and costs

Three classes (Table 3) span the fleet that is plausible before 2033. The LTV class matches the contracted rovers: 1,500 kg cargo, 10 km/h design speed modelled as a 6 km/h autonomous average, survival but not operation through the night. The 10-tonne hauler does not exist and is our proxy for the “high capacity mobility systems” NASA lists as a partnership priority [1]; its cost is an assumption built from a \$200 million vehicle plus 4 t landed at Starship's published \$100,000/kg. The small utility rover matches Lunar Outpost's HL-MAPP class. Delivered cost per vehicle combines the vehicle contract with its landing (the Phase-1 LTV task orders were about \$220 million each plus about \$234 million per Blue Moon MK1 delivery [4]).

Two cost views appear in the results. *System cost* is the fleet's amortised capital (unit cost over calendar life) plus operator hours at a loaded \$1,000 per console hour; divided by delivered mass it gives cost per kilogram moved. *Variable job cost*, used inside the auction, is the usage-driven share (50%) of a vehicle's per-operating-hour cost plus operator time. Both are assumptions and both are swept in Section 6.10.

Table 3. Vehicle classes. Sources and confidence grades for each entry are in Appendix A.

Class	Cargo (kg)	Speed, autonomous (km/h)	Night ops	Team lift	Delivered cost (\$M)	Life (yr)	MTBF / MTTR (h)	Load / unload (h)
LTV (Astrolab, Lunar Outpost class)	1,500	6.0	no	yes	450	10	500 / 24	0.5 / 0.5
HAUL (proposed heavy hauler)	10,000	4.0	yes	yes	600	10	400 / 36	1.5 / 1.5
MICRO (HL-MAPP class)	300	3.0	no	no	60	5	300 / 12	0.3 / 0.3

3.6 Scheduling and outputs

The engine keeps an event heap. Landing events create manifests and schedule READY events; READY, FREE (a vehicle finishes), NIGHT_END and WAKE events trigger a dispatch call for event-driven mechanisms; batched mechanisms act only at TICK events. When a mechanism returns jobs, the engine computes each job's duration from the slowest vehicle's approach, team set-up (1 h), handling, loaded legs, backhaul legs and charging, then adds fault recovery and night hibernation where they apply, and books operators. Delivery times for intermediate stops are interpolated along the job. The engine logs, per run, delivered and stranded mass, delays by priority and class, trips, bundled trips, team lifts, backhaul mass, kilometres and empty kilometres, energy, utilisation, operator hours, faults, night strandings, costs and revenue per owner. An item is *stranded* if no vehicle in the fleet, or no feasible team of up to three team-capable vehicles when the mechanism supports teams, can ever carry it.

4. Dispatch mechanisms

All four mechanisms see the same information at dispatch time: the waiting items that are ready and not stranded, and the vehicles that are free, alive, not on other duty and able to operate in the current lighting. None looks ahead at future landings.

In-house fleets (status quo). Every cargo owner is served by one contracted carrier: itself if it owns vehicles, otherwise a carrier assigned round-robin at the start. The carrier serves its customers' oldest item first with its nearest feasible vehicle, one item per trip. This is bilateral contracting as practised in CLPS and the LTV task orders.

Pooled greedy. Identical to in-house except that every vehicle may serve every item. This is the “closest driver” rule of ride-hailing and the simplest thing a marketplace could do.

Batched assignment. Every hour, solve a minimum-cost assignment (Hungarian algorithm [42]) between free vehicles and waiting items, with cost equal to the single-item job duration minus a small age credit so that old items are not starved. One item per vehicle. This is Uber-style batched matching and is optimal within each batch.

Marketplace (ours). Algorithm 1 gives the clearing rule. It differs from pooled greedy in six ways, each of which Section 6.8 removes in turn. (i) *Team lift*. Items heavier than any free vehicle are assigned first to the smallest team of two or three team-capable vehicles whose capacities sum to the item's mass; the team couples at the pad (1 h), drives at half speed and shares the job. (ii) *Generalised-cost bids*. For every (vehicle, item) pair the vehicle bids its variable cost for the job plus the shippers' value of the delivery time, minus an urgency credit for priority and waiting age; the globally lowest bid wins, and the auction repeats on the remaining pool (the SSI structure of [22]). (iii) *Bundling*. Before bidding, a vehicle greedily adds waiting items from the same pad while capacity allows and the detour costs less than 80% of a separate trip would; up to four distinct destinations are visited in shortest order. (iv) *Consolidation*. Deferrable bulk lots on a lane are held back until they would fill 70% of the best available vehicle or the oldest has waited 24 h, whichever comes first; a wake-up event guarantees release. (v) *Backhaul*. A vehicle finishing at a site loads any waiting return items there that fit and are bound for a pad. (vi) *Night awareness*. A vehicle without night capability refuses a job it could not finish before dark.

$$\text{bid}(v, B) = \sum_{u \in \text{team}} h(v, B) \cdot [\omega \cdot c_u + o_u \cdot p_{\text{op}}] + \sum_{i \in B} \text{VOT}(q_i) \cdot h_i(v, B) - \sum_{i \in B} [w_{q(i)} + w_{\text{age}} \cdot (t - r_i)] \cdot u \quad (1)$$

Here $h(v,B)$ is the job duration for vehicle v carrying bundle B from the current state, c_u the per-operating-hour cost of vehicle u , $\omega = 0.5$ the usage-driven wear share, o_u the operator demand (1, 1/4 or 1/20), $p_{op} = \$1,000/\text{h}$, $VOT(q)$ the value of delivery time by priority class (\$5,000, \$1,000 and \$100 per hour), h_i the time at which item i is dropped, r_i its release time, $w_1 = 6$ h, $w_2 = 2$ h, $w_3 = 0$, $w_{age} = 0.08$ h per hour waited, and $u = \$2,000$ per hour of credit. Section 6.9 searches over w_1, w_2, w_{age} and the consolidation thresholds.

Algorithm 1. Marketplace clearing (called at every READY, FREE, NIGHT_END or WAKE event)

1. **Input:** waiting items P , free vehicles F , time t .
2. **Consolidation:** for each lane of deferrable bulk lots whose total mass is below 70% of the best feasible vehicle's capacity and whose oldest lot is younger than 24 h, remove the lots from P and schedule a WAKE event at the 24 h mark.
3. **Team lift:** for each item in P heavier than every vehicle in F , in priority-then-age order, find the smallest set of 2–3 free team-capable vehicles with enough combined capacity; if the team can finish before dark (or works at night), create the job and mark the vehicles used.
4. If P has more than 40 items, keep the 40 most urgent and oldest for this call.
5. **Auction:** while unused vehicles and items remain: for every feasible (vehicle, item) pair, build a bundle by insertion and compute bid (1); take the lowest bid. If the winning vehicle cannot finish before dark, idle it and continue. Otherwise append any return items at the last stop that fit (backhaul), create the job, remove its items from P .
6. **Output:** the list of jobs. The engine prices and executes them.

Every job is paid at its bid-side variable cost, so revenue by owner is a by-product of the run; we use it only to report how evenly a pooled market spreads work across owners (Gini coefficient of revenue). A full treatment of payments, gain sharing and incentive compatibility is outside this paper; Section 7 returns to it.

5. Experimental setup

Scenarios. Table 4 lists the ten experiment suites. Each is built from the three phases by changing one or two factors. Phase 2 (2029–2032, about 17 t/yr landed plus 500 t/yr of bulk lots) is the central case because it is the first period with enough cargo and enough vehicles for dispatch to matter and the period NASA has earmarked for logistics build-out.

Replications and variance reduction. Every cell is run with 30 seeds (20 in the latency suite, which has 54 cells). The demand stream (landings, manifests, bulk lots, return items) and the operations stream (faults, duty blocks) are separate generators seeded from the cell seed, so for a given seed every mechanism faces the same cargo and the same fault history: common random numbers. We report means with 95% confidence intervals (1.96 standard errors) across seeds and, for mechanism comparisons, paired differences with their own intervals. A one-year horizon starts from an empty base; because landers arrive within days and the backlog never accumulates across lunar cycles in the reported cells, we do not discard a warm-up period.

Metrics. The primary metric is the 90th-percentile delay of urgent (priority 1) items from release to delivery, with the share of urgent items delivered within 24 h as its service-level form. Secondary metrics: delivered and stranded share of landed mass, delay by cargo class, trips per year, bundled trips, team lifts, backhaul mass, empty-kilometre share, utilisation, operator hours, faults, night strandings, system cost per kilogram and revenue Gini. Delay of deferrable bulk lots is reported separately because the marketplace delays them on purpose.

Tuning. The marketplace's hand-set weights were chosen once from a three-seed pilot and then frozen. Section 6.9 reports a random search of 40 weight configurations on Phase 2 with eight seeds and tests the winner on all phases with 30 fresh seeds. The baselines have one tunable each (the batching window and the age credit of batched assignment); we report the batching sweep in Section 6.8 and fixed the age credit at its pilot value.

Compute. All runs were made on a 12-core desktop under Windows 11. One simulated year takes 0.1–3 s depending on demand; the suite of about 7,000 runs took roughly 15 minutes on 10 worker processes. Discarded preliminary runs during development amounted to a few hundred.

Table 4. Experiment suites. Fleets are listed as owner:class; LTVs are supervised, haulers supervised or autonomous, the small rover teleoperated unless stated.

Suite	Question	Factors and levels	Mechanisms	Cells × seeds
E1 Phases	Main comparison	Phase 1 (3 vehicles), 2 (6), 3 (9) default fleets; plus an unconstrained 16-vehicle autonomous fleet as a bound	all four + bound	15 × 30
E2 Fleet size	Minimum fleet for service level	Phase-2 demand; fleets of 2, 3, 4, 5, 6, 8 vehicles	all four	24 × 30
E3 Thickness	Pooling gain vs fragmentation	6 vehicles split among 1, 2, 3, 4, 6 owners; other-duty share 0, 0.3, 0.6	in-house, marketplace	30 × 30
E4 Heavy cargo	The capability gap	Phases 2 and 3; fleets of 4 LTVs, 3 LTVs + 1 hauler, 2 LTVs + 2 haulers	greedy, marketplace, marketplace without team lift	18 × 30
E5 Autonomy and latency	Throughput bounds	All-teleop, all-supervised, all-autonomous 5-vehicle fleets; RTT 2.6, 5, 8, 16, 603, 1,323 s; 1, 2, 4 operators	marketplace	54 × 20
E6 Standards	Interface compatibility	Share of cargo on the common standard 0.5, 0.75, 0.9, 1.0; 3 of 5 vehicles accept only the common standard	in-house, greedy, marketplace	12 × 30
E7 Environment	Robustness	Dark share 0.1–0.5; MTBF × 0.25–×2; bulk 0–3,000 t/yr	greedy, marketplace	24 × 30
E8 Ablations	Which components matter	Each of six components removed; batching window 0, 0.5, 1, 2, 6 h; Phases 2 and 3 with tight fleets	marketplace variants	22 × 30
E9 Tuning	Do tuned weights transfer?	40 random weight configurations on Phase 2 (8 seeds); best vs hand-set on Phases 1–3 (30 seeds)	marketplace	41 × 8 + 6 × 30
E10 Economics	Cost drivers	Demand × 0.5/×1.5; operator cost; other duty; dark share; MTBF; plus analytic vehicle cost and life	marketplace	11 × 30

6. Results

6.1 Main comparison across phases

Pooling of any kind beats bilateral contracting, in every phase. Figure 4 and Table 5 give the main comparison with each phase's default fleet (3, 6 and 9 vehicles). In-house fleets deliver urgent

cargo with a 90th-percentile delay of 43 h in Phase 1, 40 h in Phase 2 and 20 h in Phase 3; the marketplace brings those to 26, 16 and 6.5 h, paired differences of -17.6 ± 8.4 , -24.7 ± 9.5 and -13.4 ± 4.8 h (95% CI over 30 seeds). The share of urgent items delivered within a day rises by 15, 12 and 7 absolute points. The in-house rule also leaves 6–13 tonnes of landed cargo undelivered at year end in Phases 2 and 3, because items heavier than 1.5 t belong to owners whose contracted carrier has no hauler while another company's hauler sits idle.

Among pooled rules, urgent-cargo delay is nearly the same. Pooled greedy, batched assignment and the marketplace are within a few hours of each other. The marketplace's advantage over pooled greedy on the primary metric is -4.5 ± 4.0 h in Phase 2 (a 23% relative reduction) and not distinguishable from zero in Phases 1 and 3; against batched assignment it is -0.8 ± 0.9 , -2.5 ± 3.8 and -0.4 ± 0.7 h. The unconstrained 16-vehicle autonomous fleet delivers urgent items in 2.1–2.4 h on average, which is essentially travel and handling time. Default fleets therefore spend most of the urgent delay waiting for a vehicle to be free, lit and off other duty, not driving.

Where the marketplace differs is in what it can move and when it drives. With LTV-class rovers and one hauler, 9–14% of landed mass in every phase is heavier than any single vehicle. The marketplace's team lifts cut that to 0% in Phases 1 and 3 and 4.2% in Phase 2 (the residual is 13–15 t elements that even a hauler plus two LTVs cannot carry together). Vehicles without night capability are caught by the lunar night 0.2, 1.4 and 4.7 times a year under greedy dispatch and 0.03, 0.07 and 0.17 times under the marketplace, each event costing a job about 106 h. Bundling cuts trips by 4–7%. The price is a deliberate one: deferrable bulk lots wait 17–35 h on average under consolidation against 3–13 h under greedy dispatch. Operator hours fall slightly in Phases 1 and 2 and rise by 223 ± 24 h (28%) in Phase 3, where the generalised-cost bid lets cheap, slow, remotely operated vehicles win bulk lots; Section 6.9 shows that tuned weights remove this increase.

Pooling also spreads the work. The Gini coefficient of revenue across vehicle owners is 0.46–0.72 under in-house contracting and 0.20–0.47 under the pooled rules. Empty kilometres are about half of all kilometres under every rule, far above the 16–21% deadhead of Earth trucking, because almost all cargo flows one way, from pads to base; return items are too few to change this.

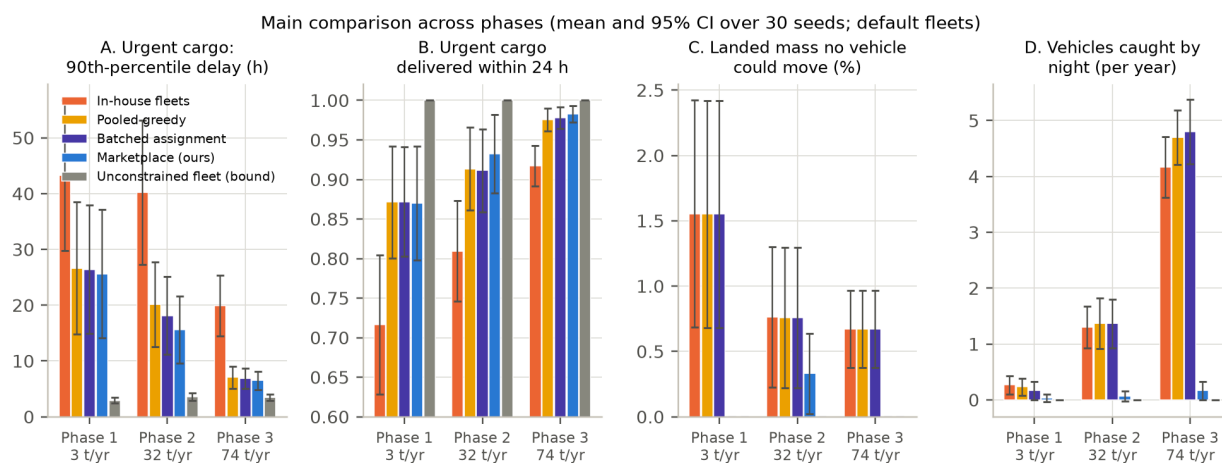


Figure 4. Main comparison with each phase's default fleet. Bars are means over 30 seeds; whiskers are 95% confidence intervals. **A:** 90th-percentile delay of urgent items from release to delivery. **B:** share of urgent items delivered within 24 h. **C:** share of landed mass that no vehicle or team in the fleet could ever move. **D:** number of times per year a vehicle without night capability is caught out by the lunar night. The grey bars are a 16-vehicle fully autonomous fleet with no other duties or faults, included as a bound.

Table 5. Main results with default fleets: mean $\pm$ 95% CI over 30 seeds. Urgent = priority-1 items. Immovable = share of landed cargo mass no vehicle or team could ever carry. Bulk delay is the mean delay of deferrable regolith lots, which the marketplace holds on purpose. Cost is amortised fleet plus operator time per kilogram of all mass moved.

Phase	Mechanism	Urgent p90 delay (h)	Urgent mean delay (h)	Urgent within 24 h	Immovable (%)	Night strandings / yr	Trips / yr	Bulk delay (h)	Operator h / yr	Revenue Gini	Cost \$/kg moved
Phase 1	In-house fleets	43.3 $\pm$ 13.4	19.4 $\pm$ 6.2	0.72 $\pm$ 0.09	13.7	0.27 $\pm$ 0.16	74	19.7	52	0.46	3,105
Phase 1	Pooled greedy	26.6 $\pm$ 11.8	12.0 $\pm$ 5.5	0.87 $\pm$ 0.07	13.7	0.23 $\pm$ 0.15	75	13.1	54	0.35	3,084
Phase 1	Batched assignment	26.4 $\pm$ 11.5	12.2 $\pm$ 5.4	0.87 $\pm$ 0.07	13.7	0.17 $\pm$ 0.17	75	13.2	51	0.35	3,084
Phase 1	Marketplace (ours)	25.6 $\pm$ 11.5	11.6 $\pm$ 5.5	0.87 $\pm$ 0.07	0.0	0.03 $\pm$ 0.07	72	34.7	44	0.35	3,034
Phase 1	Unconstrained 16-vehicle bound	2.9 $\pm$ 0.5	2.1 $\pm$ 0.2	1.00 $\pm$ 0.00	0.0	0.00 $\pm$ 0.00	74	26.4	7	0.00	23,191
Phase 2	In-house fleets	40.2 $\pm$ 13.0	15.6 $\pm$ 3.9	0.81 $\pm$ 0.06	9.4	1.30 $\pm$ 0.38	549	12.3	364	0.67	489
Phase 2	Pooled greedy	20.1 $\pm$ 7.6	7.8 $\pm$ 2.3	0.91 $\pm$ 0.05	9.4	1.37 $\pm$ 0.46	555	4.3	495	0.47	476
Phase 2	Batched assignment	18.1 $\pm$ 7.0	8.2 $\pm$ 2.2	0.91 $\pm$ 0.05	9.4	1.37 $\pm$ 0.44	555	4.5	440	0.44	476
Phase 2	Marketplace (ours)	15.6 $\pm$ 6.0	6.5 $\pm$ 2.0	0.93 $\pm$ 0.05	4.2	0.07 $\pm$ 0.09	531	23.5	456	0.45	474
Phase 2	Unconstrained 16-vehicle bound	3.6 $\pm$ 0.7	2.4 $\pm$ 0.3	1.00 $\pm$ 0.00	0.0	0.00 $\pm$ 0.00	544	22.3	55	0.00	1,458
Phase 3	In-house fleets	19.9 $\pm$ 5.4	8.3 $\pm$ 1.6	0.92 $\pm$ 0.03	12.5	4.17 $\pm$ 0.55	1693	7.4	908	0.72	267
Phase 3	Pooled greedy	7.0 $\pm$ 2.0	4.2 $\pm$ 0.8	0.98 $\pm$ 0.01	12.5	4.70 $\pm$ 0.49	1696	3.2	795	0.22	266
Phase 3	Batched assignment	6.9 $\pm$ 1.8	4.4 $\pm$ 0.7	0.98 $\pm$ 0.01	12.5	4.80 $\pm$ 0.57	1696	3.6	781	0.23	266
Phase 3	Marketplace (ours)	6.5 $\pm$ 1.6	3.6 $\pm$ 0.5	0.98 $\pm$ 0.01	0.0	0.17 $\pm$ 0.17	1572	16.9	1018	0.20	265
Phase 3	Unconstrained 16-vehicle bound	3.5 $\pm$ 0.5	2.2 $\pm$ 0.1	1.00 $\pm$ 0.00	0.0	0.00 $\pm$ 0.00	1607	16.1	181	0.00	494

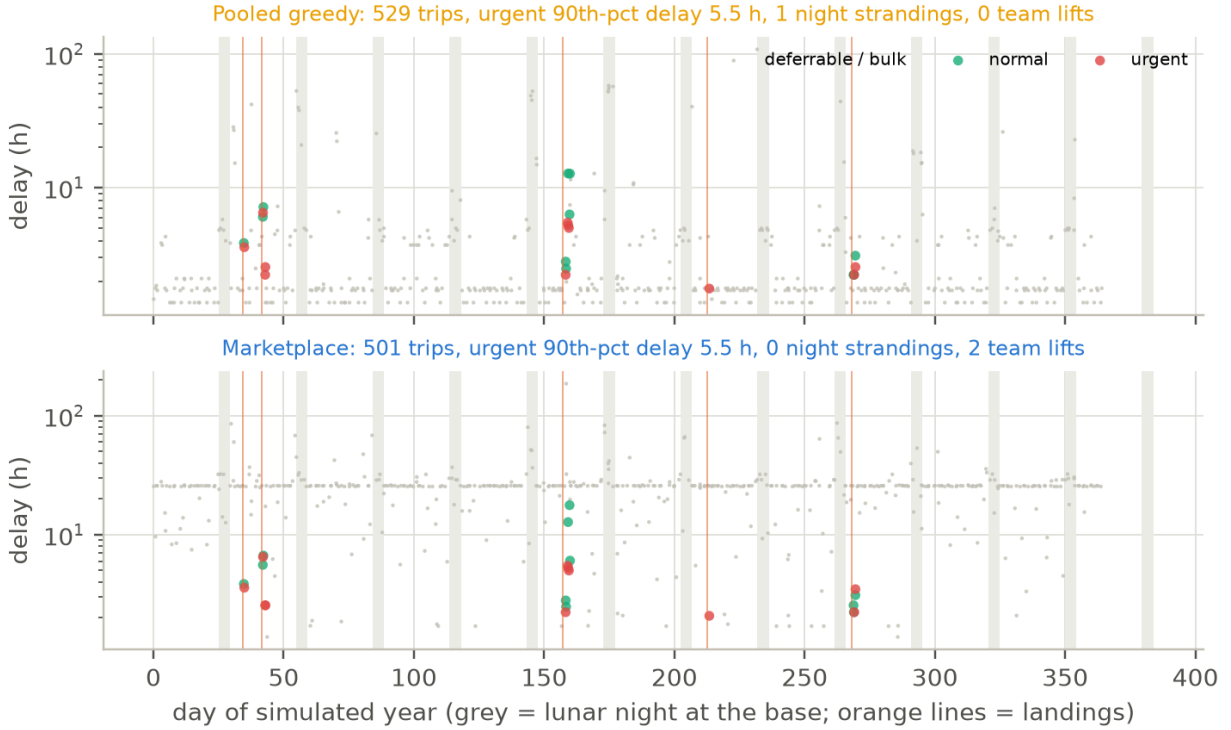


Figure 5. One simulated Phase-2 year (seed 3, four-vehicle fleet) under pooled greedy dispatch (top) and the marketplace (bottom). Each dot is a delivered item at its release day and its delay. Grey bands are lunar nights at the base, orange lines are landings. The marketplace's deferrable lots cluster at the 24 h consolidation limit; its urgent items (red) are served at the same speed as under greedy dispatch, and no vehicle is caught by the night.

6.2 How small a fleet can serve Phase 2?

Because the fleet is the cost, the practical question is not which rule delivers fastest with a given fleet, but which rule meets a service target with the fewest vehicles. Figure 6 sweeps fleets of 2 to 8 vehicles at Phase-2 demand. The marketplace and batched assignment reach a 90% same-day service level for urgent cargo with 4 vehicles (0.895 ± 0.061 and 0.900 ± 0.063); pooled greedy reaches it with 6 (0.914 ; 0.890 with 5); in-house fleets peak at 0.84 with 8. In annual fleet cost at our unit prices, 4 vehicles are \$162 million a year and 6 are \$252 million, so the dispatch rule alone is worth about \$90 million a year, or 36% of the fleet bill, at this demand level. Below 4 vehicles no rule reaches the target: with two LTVs the 90th-percentile urgent delay is 45–53 h for the pooled rules and 80 h in-house, because a single landing can release more items than two rovers can clear before the next night.

System cost per kilogram of all mass moved (including regolith) rises almost linearly with fleet size, from about \$175 with 2 vehicles to about \$670 with 8, because demand is fixed and the fleet is 99% of the cost. Per kilogram of *landed* cargo the picture depends on what gets moved: with two LTVs the marketplace delivers heavy items by team lift that greedy dispatch leaves stranded, so its landed-kg cost is \$3,000 against \$11,700 for greedy; with four vehicles the rules are within \$11,000–15,000 per landed kilogram at Phase-2 volumes, and the figure rises with every vehicle added.

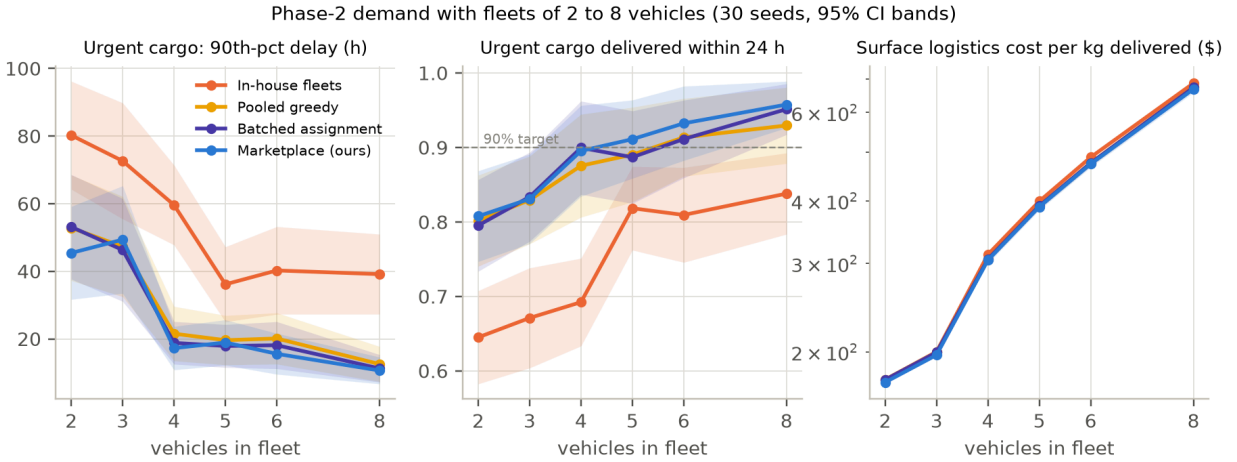


Figure 6. Fleet-size sweep at Phase-2 demand (30 seeds; bands are 95% CIs). Fleets grow from two LTVs to eight mixed vehicles (Table 4). **Left:** urgent 90th-percentile delay. **Middle:** share of urgent items delivered within 24 h with the 90% target marked. **Right:** system cost per kilogram of all mass moved; the curves lie on top of each other because the fleet, not the dispatch rule, sets the cost.

6.3 Market thickness: when does pooling pay?

Figure 7 holds the fleet fixed at six vehicles and splits it among one to six owners, with each owner's cargo contracted to its own vehicles under the in-house rule. With a single owner the in-house rule and the marketplace are statistically indistinguishable: the paired gain in urgent 90th-percentile delay is 3.3 ± 3.2 h with no other duties, 2.2 ± 3.6 h at 30% other duties and 0.9 ± 5.5 h at 60%. The gain grows monotonically with fragmentation. With six owners it is 23 ± 9 h, 40 ± 9 h and 54 ± 13 h at the three duty levels; the in-house same-day service level falls from 0.93 to 0.87, 0.70 and 0.53 while the marketplace's stays at 0.95, 0.93 and 0.89 regardless of how many owners there are. Other duties amplify the effect because a vehicle committed to its owner's science traverse cannot serve its owner's cargo, while under pooling a neighbour's idle vehicle can.

This is the thin-market result we expected from Earth freight, but in a sharper form. The lunar market will have very few participants, so whether pooling matters depends almost entirely on how many of them there are. If NASA's two LTV providers and one or two commercial payload companies each run their own vehicles for their own cargo, the simulations say that a shared dispatch layer roughly halves urgent-cargo delay and keeps every owner's heavy items moving; if a single integrator owns the whole fleet, the dispatch layer adds little that a good in-house scheduler would not.

Market thickness: pooling gain grows with fragmentation and with vehicles' other duties

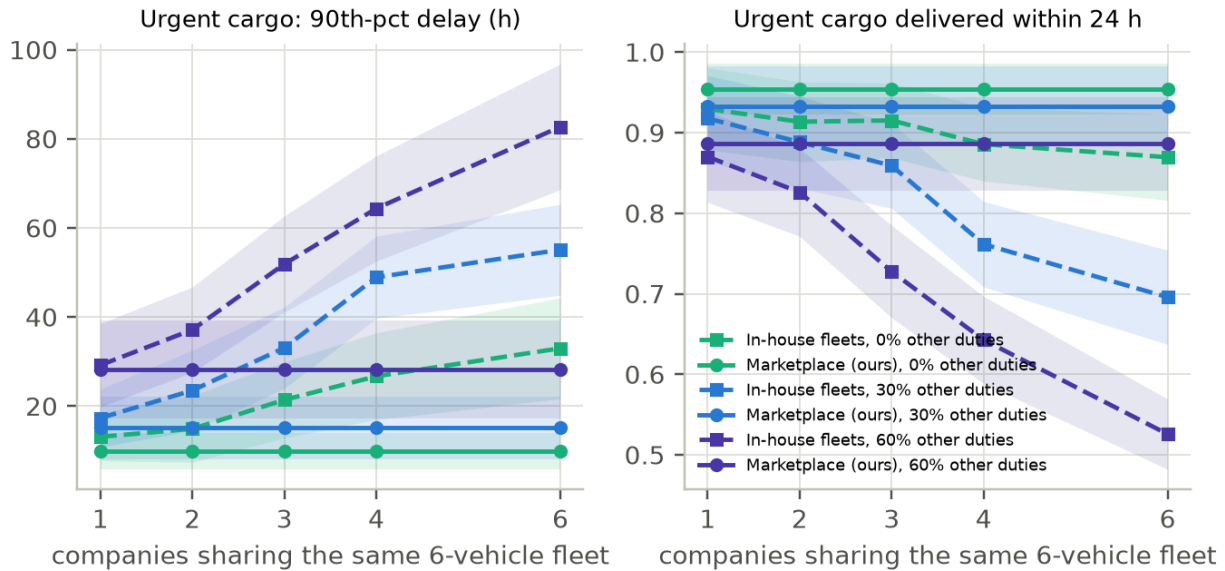


Figure 7. Market thickness. Six vehicles (five LTVs and one hauler) are split among 1, 2, 3, 4 or 6 owners; cargo is drawn from the same owners. Dashed lines with squares are in-house contracting, solid lines with circles the marketplace; colours give the share of time vehicles spend on their owners' other duties. Bands are 95% CIs over 30 seeds. The marketplace lines are flat because pooling makes ownership irrelevant to dispatch.

6.4 The heavy-cargo gap

NASA's statement that no planned vehicle can relocate a large element translates, in the simulations, into a large share of immovable mass. With four LTV-class rovers and no hauler, $54 \pm 8\%$ of landed cargo mass in Phase 2 and $63 \pm 7\%$ in Phase 3 is heavier than any vehicle (Figure 8). Team lifts by two or three LTVs, which can carry up to 3 and 4.5 t together, recover about half of that: the immovable share falls to $29 \pm 9\%$ and $38 \pm 7\%$. Replacing one LTV with a 10-tonne hauler leaves 9% and 12.5% immovable under greedy dispatch (the 12–15 t elements), and 4.2% and 2.7% under the marketplace, where a hauler and two LTVs lift together. Two haulers remove the gap entirely in both phases. Items that *can* be moved are not slowed by the team lifts: the urgent 90th-percentile delay with one hauler is 15.8 h under the marketplace against 20.3 h under greedy dispatch in Phase 2.

We read this as follows. Coordination software can stretch a fleet of 1.5-tonne rovers to the 2–3 tonne items that MK1-class landers deliver, which is most of the Phase-2 manifest by count. It cannot reach the 6–15 tonne elements that define Phase 3. One heavy vehicle, dispatched as part of a shared pool so that it also lends its capacity to others' items, closes most of the remaining gap; a second closes all of it.

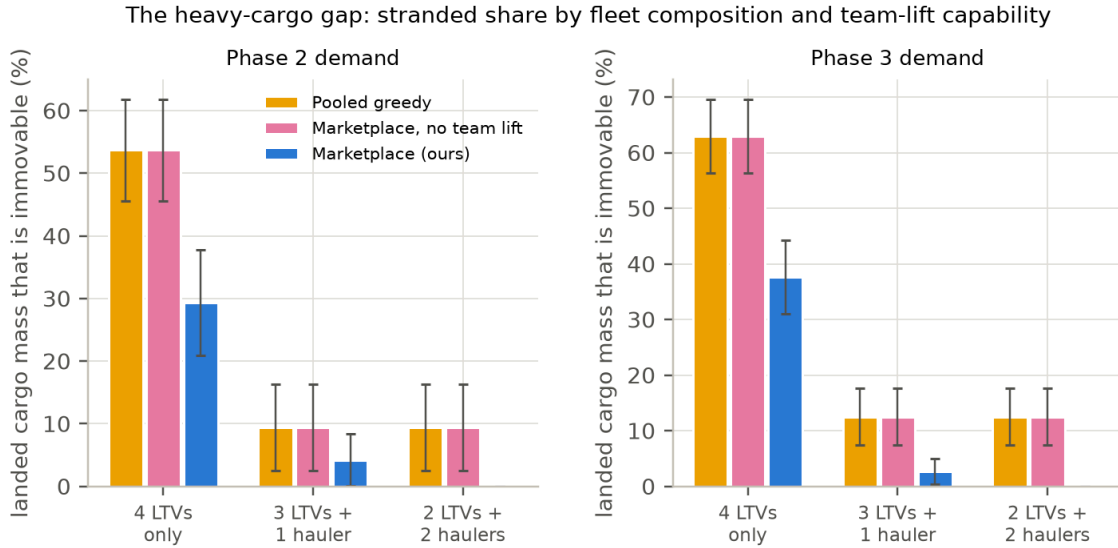


Figure 8. Share of landed cargo mass that no vehicle or team could move, by fleet composition and phase (30 seeds; 95% CIs). Team lift (blue) recovers about half of the mass stranded by an LTV-only fleet and most of the mass stranded with a single hauler.

6.5 Autonomy and latency

Figure 9 compares a five-vehicle Phase-2 fleet operated entirely by Earth teleoperation, supervised autonomy or full autonomy, across round-trip latencies from 2.6 s (direct light time) to 1,323 s (Mars at its farthest). With supervised or full autonomy the latency does not matter in our model: urgent 90th-percentile delay is 17 and 16 h at every latency, and operator hours are 344 and 63 per year. Teleoperation is workable at lunar latency but costly: with two operators the urgent 90th-percentile delay is 39 h at 2.6 s, 55 h at the 5 s default and 60 h at 8 s, rising to 127 h at a relayed 16 s; operator consoles are staffed 2,300–6,000 hours a year, that is one to three full-time positions for five rovers. A single operator for five teleoperated rovers is a bottleneck in its own right: jobs wait for an operator 280–580 times a year and the 90th-percentile delay doubles relative to two operators. At Mars-like latencies teleoperated speed falls below 0.05 km/h and the fleet delivers only 3–15% of the year's mass.

Two conclusions follow for a marketplace that dispatches other people's vehicles. First, the vehicles' autonomy level is a market-design variable: a pool of teleoperated rovers needs the marketplace to schedule operators as well as vehicles, and even then runs 2–3 times slower on urgent cargo. Second, nothing that works by teleoperation on the Moon transfers to Mars; the Mars column in Figure 9 is a different problem.

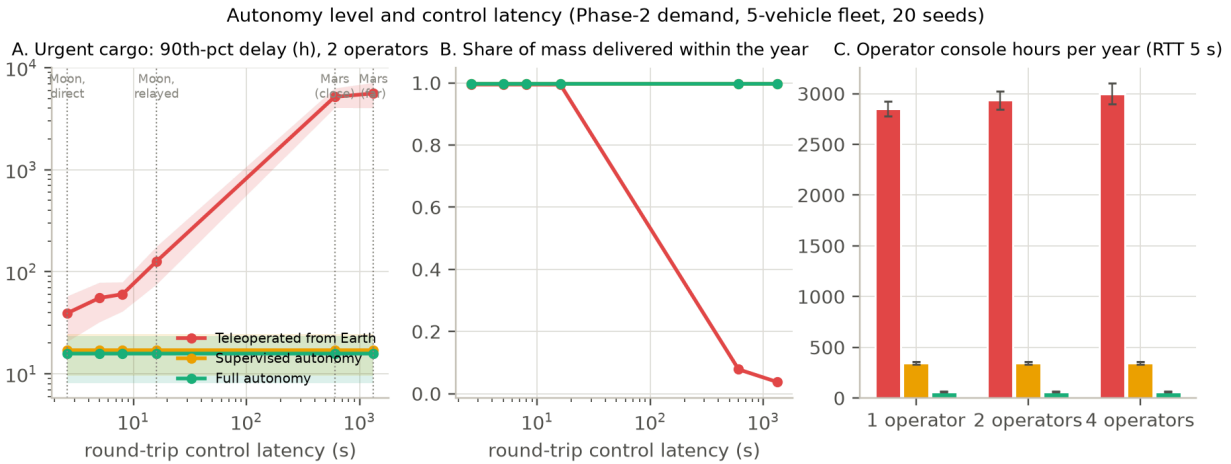


Figure 9. Autonomy level and control latency at Phase-2 demand with a five-vehicle fleet (20 seeds; bands are 95% CIs). **A:** urgent 90th-percentile delay with two Earth operators, on log axes; the dotted lines mark direct and relayed lunar latency and the near and far Mars cases. **B:** share of the year's mass delivered. **C:** operator console hours per year at the 5 s default latency for one, two and four operators.

6.6 Interface standards

In this suite three of five vehicles accept only the common cargo standard and two accept both, while the share of landed cargo built to the common standard varies from 50% to 100% (Figure 10). Nothing becomes immovable, because every item has at least two compatible vehicles, but delays grow as compatibility shrinks: at 50% common-standard cargo the in-house urgent 90th-percentile delay is 49 h against 36 h at full compatibility, and the marketplace's is 22 h against 19 h. The service level for urgent items moves from 0.76 to 0.82 in-house and from 0.89 to 0.91 under the marketplace. Pooling therefore does most of the work that a universal standard would do here, because it lets the two flexible vehicles serve everybody's non-standard items. The remaining gap is small in this fleet; it would be large in a fleet where only one company's vehicles accept a second standard.

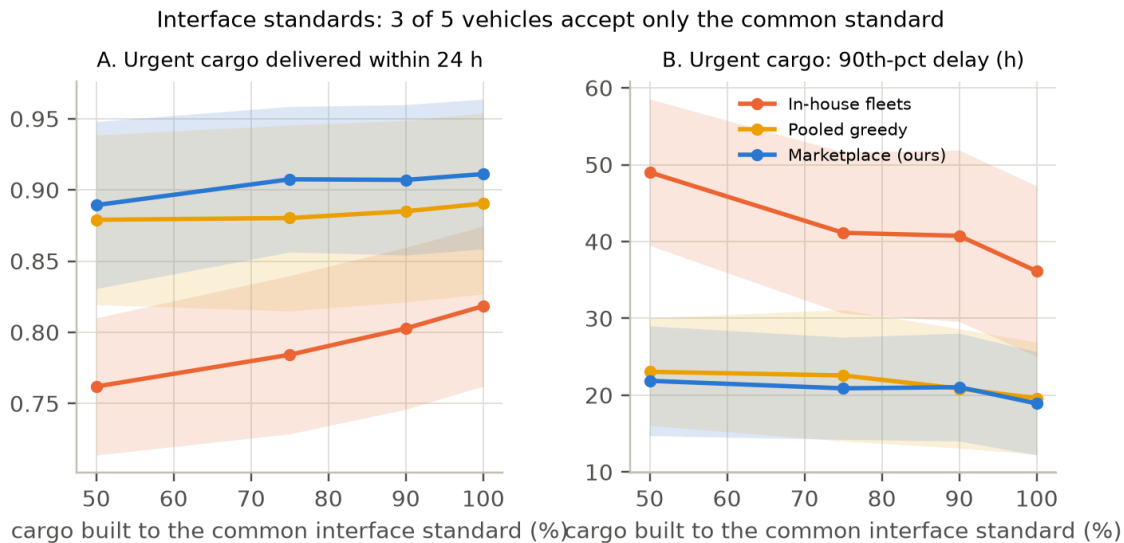


Figure 10. Interface standards at Phase-2 demand (30 seeds; 95% CIs). Three of five vehicles accept only the common standard. **A:** urgent same-day service level. **B:** urgent 90th-percentile delay.

6.7 Night, reliability and bulk flows

Figure 11 sweeps three environment factors with a five-vehicle Phase-2 fleet. **Night.** Delay grows steeply with the dark share of the cycle: at 30% dark the urgent 90th-percentile delay is 47 h under greedy dispatch and 31 h under the marketplace (paired difference -15 ± 9 h); at 50% dark, 72 against 50 h (-23 ± 11 h). The difference is night awareness: greedy dispatch sends vehicles out that cannot return before dark 1–1.5 times a year regardless of the dark share, and each such event removes the vehicle and its cargo for the whole night. **Reliability.** Quartering the MTBF (to about 125 h for an LTV, 11 faults a year across the fleet) raises the greedy 90th-percentile delay from 20 to 26 h and the marketplace's from 19 to 21 h; the paired difference, -5.6 ± 6.7 h, is not distinguishable from zero. Faults matter less than nights because recovery takes 12–36 h rather than 106. **Bulk flows.** Raising regolith and feedstock flows from zero to 3,000 t a year raises utilisation from 0.5% to 16–17% and leaves urgent delay roughly unchanged until the top level, where both rules degrade to about 31–33 h; the marketplace makes 5–14% fewer trips, more as the flows grow, and the greedy rule's night strandings rise to 6.4 a year. Even at 3,000 t a year the fleet is idle five sixths of the time, which is a reminder that these vehicles will be bought for capability, not for throughput.

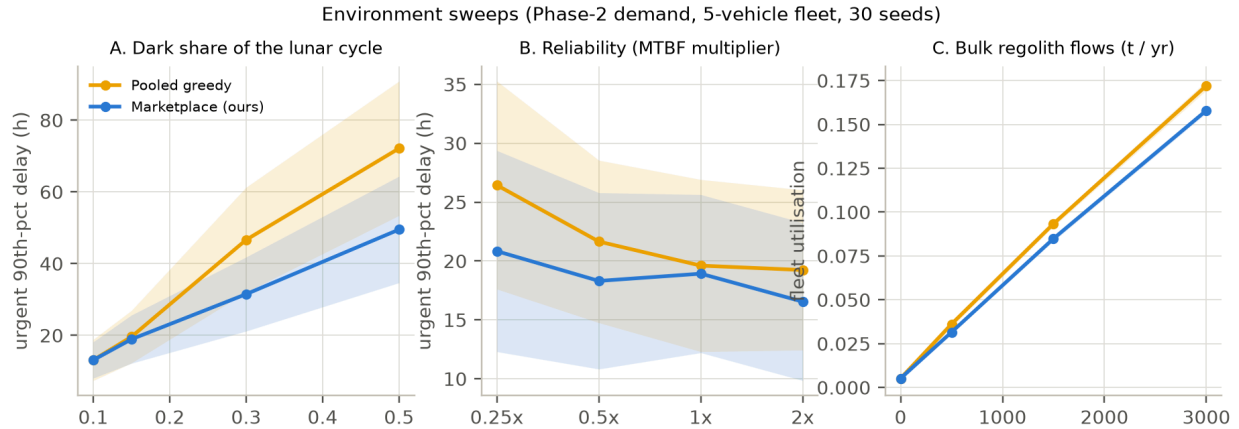


Figure 11. Environment sweeps at Phase-2 demand with a five-vehicle fleet (30 seeds; bands are 95% CIs). **A:** dark share of the lunar cycle at the base. **B:** mean time between faults as a multiple of the default. **C:** bulk regolith and feedstock flows, shown as fleet utilisation.

6.8 Ablations and the batching window

Figure 12 and Table 6 remove one marketplace component at a time with tight fleets (four vehicles in Phase 2, six in Phase 3). Two components carry the urgent-delay result: without bundling the 90th-percentile delay rises by 2.8 ± 2.7 h in Phase 2 and 4.6 ± 4.1 h in Phase 3, with 31 and 129 more trips a year; without consolidation it rises 3.6 ± 2.8 h in Phase 3 with 61 more trips. Night awareness carries the night result: removing it brings back 1.1 ± 0.3 and 5.4 ± 1.0 night strandings a year and adds 40 ± 15 and 242 ± 55 operator hours (hibernating vehicles keep their operators booked). Team lift carries the stranded-mass result and costs nothing on urgent delay; in Phase 3 its removal even improves urgent delay by 2.0 ± 1.7 h because vehicles are no longer tied up in pairs. Two components did nothing measurable in these scenarios: the priority credit (-0.2 ± 0.4 and -0.3 ± 0.8 h) and backhaul (0.0 and 0.1 ± 0.2 h). The value-of-time term in the bid already orders urgent items first, and return items are rare enough that a vehicle seldom finds one at its last stop. We keep both in the mechanism for completeness but make no claim for them.

The batching window behaves as the thin-market theory predicts [11]. Dispatching on every event is best or tied in both phases; windows of 0.5–2 h add 1–5 h to the urgent 90th-percentile delay, and a 6 h window adds 9 h in Phase 2 and 21 h in Phase 3 while bundling only 10–17 more trips. There is nothing to gain from waiting when the next vehicle to free up is hours away and the next item to arrive is days away.

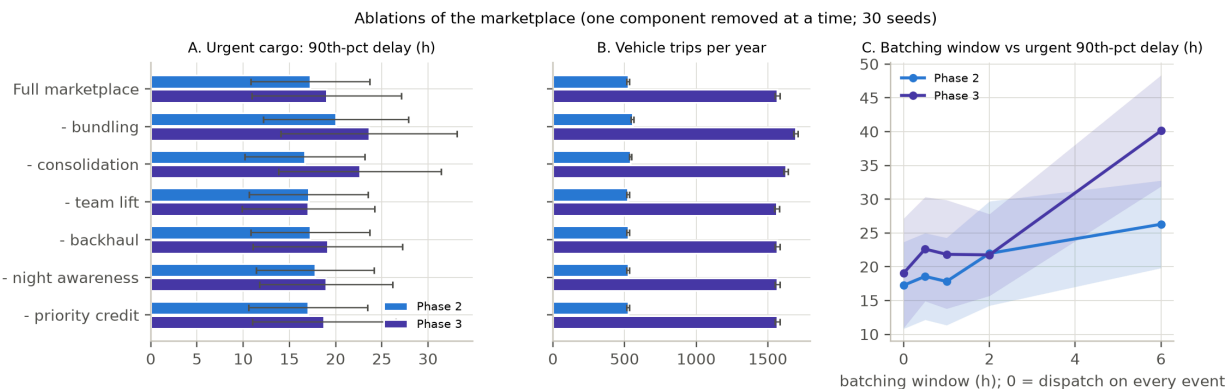


Figure 12. Ablations with tight fleets (four vehicles in Phase 2, six in Phase 3; 30 seeds; 95% CIs). **A:** urgent 90th-percentile delay with each component removed. **B:** vehicle trips per year. **C:** urgent delay as a function of the batching window; zero means dispatch on every event.

Table 6. Ablations with tight fleets (Phase 2: four vehicles; Phase 3: six vehicles). Each cell is the mean over 30 seeds; the bracketed value is the paired difference from the full marketplace with its 95% CI. Immovable = share of landed cargo mass no vehicle or team could carry.

Phase	Variant	Urgent p90 delay (h)	Trips / yr	Night strandings / yr	Immovable (%)	Bulk delay (h)	Operator h / yr
Phase 2	Full marketplace	17.3	524	0.03	4.2	24.7	406
Phase 2	without bundling	20.0 [+2.8 ± 2.7]	555 [+31 ± 3]	0.07 [+0.03 ± 0.11]	4.2	4.4 [-20.3 ± 0.2]	408 [+2 ± 9]
Phase 2	without consolidation	16.7 [-0.6 ± 2.0]	541 [+17 ± 2]	0.03 [+0.00 ± 0.00]	4.2	4.4 [-20.3 ± 0.2]	397 [-8 ± 6]
Phase 2	without team lift	17.1 [-0.2 ± 0.3]	523 [-0 ± 1]	0.03 [+0.00 ± 0.00]	9.4	24.7 [+0.0 ± 0.0]	402 [-4 ± 4]
Phase 2	without backhaul	17.3 [+0.0 ± 0.0]	524 [+0 ± 0]	0.03 [+0.00 ± 0.00]	4.2	24.7 [-0.0 ± 0.0]	406 [+0 ± 0]
Phase 2	without night awareness	17.8 [+0.5 ± 0.6]	524 [+0 ± 1]	1.13 [+1.10 ± 0.34]	4.2	24.9 [+0.1 ± 0.1]	446 [+40 ± 15]
Phase 2	without priority credit	17.0 [-0.2 ± 0.4]	524 [+0 ± 0]	0.03 [+0.00 ± 0.00]	4.2	24.7 [-0.0 ± 0.0]	405 [-1 ± 1]
Phase 3	Full marketplace	19.0	1566	0.30	2.7	17.9	1846
Phase 3	without bundling	23.6 [+4.6 ± 4.1]	1695 [+129 ± 6]	0.27 [-0.03 ± 0.32]	2.7	5.8 [-12.1 ± 0.2]	1511 [-334 ± 23]
Phase 3	without consolidation	22.6 [+3.6 ± 2.8]	1626 [+61 ± 7]	0.27 [-0.03 ± 0.32]	2.7	6.1 [-11.8 ± 0.2]	1477 [-368 ± 23]
Phase 3	without team lift	17.0 [-2.0 ± 1.7]	1563 [-2 ± 3]	0.20 [-0.10 ± 0.24]	12.5	17.8 [-0.1 ± 0.1]	1800 [-45 ± 21]
Phase 3	without backhaul	19.1 [+0.1 ± 0.2]	1566 [+1 ± 1]	0.30 [+0.00 ± 0.09]	2.7	17.9 [-0.0 ± 0.0]	1847 [+2 ± 10]
Phase 3	without night awareness	19.0 [-0.1 ± 2.3]	1567 [+1 ± 3]	5.73 [+5.43 ± 1.04]	2.7	18.1 [+0.2 ± 0.1]	2088 [+242 ± 55]
Phase 3	without priority credit	18.7 [-0.3 ± 0.8]	1567 [+1 ± 2]	0.33 [+0.03 ± 0.22]	2.7	17.8 [-0.1 ± 0.1]	1847 [+1 ± 19]

6.9 Do tuned weights transfer?

We searched 40 random weight configurations on the Phase-2 tight-fleet scenario with eight seeds and tested the best against the hand-set weights on all three phases with 30 fresh seeds (Figure 13). Across the 40 configurations the urgent 90th-percentile delay on the search set ranged from 9.7 to 15.9 h with a mean of 11.8 h; the hand-set configuration scored 10.1 h, in the top quarter. The best configuration uses a much larger urgency credit for priority-1 items (22.5 h against 6 h), a smaller one for priority 2, and releases bulk lots sooner (32% fill or 12 h, against 70% or 24 h). On the test seeds it improves the urgent 90th-percentile delay by 2.0 ± 2.3 h in Phase 2 and 3.0 ± 5.2 h in Phase 3, neither distinguishable from zero, and leaves Phase 1 unchanged. What it does change is the side costs: bulk lots wait 6–22 h less and Phase-3 operator hours fall by 219 ± 25 , which removes the operator-hour penalty seen in Section 6.1. We take two things from this. The mechanism's urgent-cargo performance is set by its structure (pooling, team lift, night awareness, event-driven dispatch) rather than by its weights; and the weights that matter trade bulk-lot patience against operator time, which is a policy choice an operator can make per base rather than a parameter to fit.

Auction weights searched on Phase 2 (40 random configs x 8 seeds), tested on all phases with 30 fresh seeds. Best: $w_1=22.5$, $w_2=0.3$, $age=0.19$, $fill=0.32$, $wait=12$ h

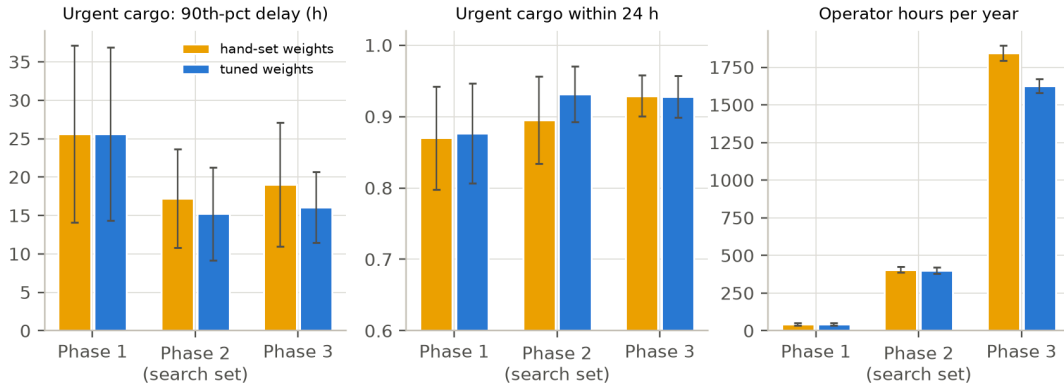


Figure 13. Hand-set versus tuned auction weights (30 fresh seeds per phase; 95% CIs). Weights were searched on Phase 2 only. Urgent-cargo metrics are unchanged within confidence intervals; the tuned weights reduce operator hours in Phase 3 and shorten bulk-lot waits in every phase.

6.10 What a surface move costs

Figure 14 puts the cost side together for the Phase-2 four-vehicle marketplace cell. The annual bill is \$162 million of amortised fleet and \$0.4 million of operator time. Divided by the 534 t moved in the year (32 t of landed cargo and 500 t of regolith and feedstock) that is \$305 per kilogram moved; divided by landed cargo alone it is \$4,800 per kilogram, and \$5,200 in Phase 3 with the nine-vehicle default fleet. The tornado in panel A shows that only two assumptions move this number: vehicle unit cost ($\pm 50\%$ gives \$152–456 per kg) and vehicle life (15 years or 6.7 years gives \$203–456). Halving or multiplying demand by 1.5 changes it by 2%; operator cost by 1%; other duties, dark share and reliability by nothing, because they change delay, not cost. Panel B puts the surface move next to the cost of getting the kilogram to the surface: about \$1.08 million per kg on a CLPS small lander (Blue Ghost Mission 1), about \$78,000 on a fully loaded Blue Moon MK1 and SpaceX's published \$100,000 for Starship. Moving a kilogram across the base costs 0.5–6% of landing it.

Three things follow for anyone pricing the service. The marginal cost of a job is small and the fixed cost is enormous, so prices will be set by capacity commitments rather than by job costs, as in contract trucking rather than the spot market. Dispatch quality shows up as fleet size, not operating cost: the \$90 million-a-year difference between four and six vehicles in Section 6.2 is the economic content of the mechanism comparison. And the service is cheap relative to what it protects: a day's delay on a \$1 million-per-kilogram crate is worth far more than the few thousand dollars it costs to move it, which is why urgent-cargo delay rather than cost is the right primary metric for this market.

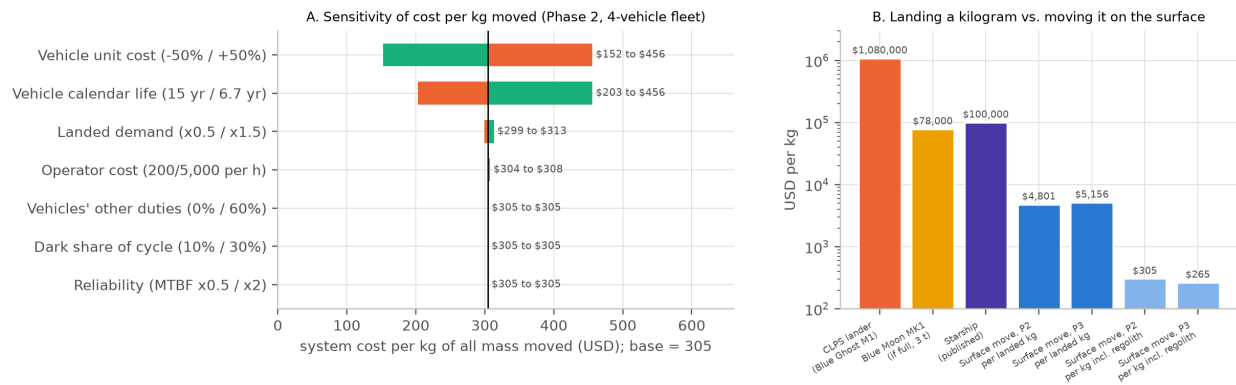


Figure 14. Economics. **A:** tornado of system cost per kilogram moved for the Phase-2 four-vehicle marketplace cell; vehicle unit cost and life are analytic swings on the cost model, the others are simulated (30 seeds). **B:** cost of landing a kilogram on three vehicles [46, 47, 4] against the simulated cost of moving it on the surface, per landed kilogram and per kilogram of all mass moved including regolith.

7. Discussion

7.1 What the results say to a marketplace operator

The case for a neutral dispatch layer on the Moon rests on three findings, and each comes with a condition. Pooling helps only when the fleet is fragmented across owners, and the market NASA is building is fragmented by design: two LTV providers who may hire out their rovers between NASA missions, several payload companies, international partners, and an agency that has said it will not own the vehicles. The mechanism's value shows up as fleet size: at Phase-2 demand it holds the same service level with four vehicles instead of six, which at today's prices is worth about \$90 million a year, and it keeps every owner's heavy items moving instead of leaving the only hauler idle while another company's habitat waits. And the components that carry those results are mundane: pool everybody's vehicles, dispatch on every event, bundle items from the same pad, hold deferrable lots for half a day, never send a rover out that cannot get home before dark, and let vehicles lift together. The auction weights barely matter.

Those conditions shape the business. The customer base will be small for a decade, and the anchor customer will be NASA; the Earth analogue is contract trucking with a thin spot layer, not a ride-hailing app [27]. Revenue scales with fleet capital, not with job counts, so a platform fee of a few percent of the fleet's annual cost is the plausible unit: on a \$162 million-a-year Phase-2 fleet that is a business of a few million dollars a year, which is small but real and grows with each phase. The platform's technical core is the dispatch rule, the operator scheduler and the data model (manifests, interfaces, organics inventories for shadowed sites, safety-zone geometry under the Artemis Accords [51]); the export-control status of rover command software [52] decides who may run it and from where.

For Roatify specifically, the results support the staged plan we set out before this study: build and run the dispatch platform on Earth freight now, where data and customers exist; test the lunar version in simulation with NASA's published numbers and with rover makers, which is what this paper does and what NASA's NextSTEP-3 demonstrations require at TRL 6; and launch the surface marketplace when Phase 2 brings enough vehicles and owners for pooling to matter, which the simulations place at three or more owners. The later ambition of building a heavy hauler is also sharpened: the simulations say the hauler is what Phase 3 needs and software cannot replace it, but a single shared hauler does most of the work of two private ones.

7.2 What the results say to NASA

Three of NASA's stated gaps appear in the results with numbers attached. "Multiple robotic mobility systems working together" recovers about half of the mass an LTV-only fleet cannot move. "High capacity mobility systems" are the only way to move the other half, and one such vehicle in a shared pool is worth two in private hands. "Autonomy or tele-robotic operation throughout the year" is a factor-of-two to factor-of-three lever on urgent delay and a factor of seven to thirty on operator hours. A fourth gap is implicit in the thickness result: a shared dispatch layer is what makes the agency's intention to let LTV providers serve other customers produce a service rather than two private taxis.

7.3 What we would build next, and what it would cost

The testbed answers the questions it was built for and leaves three undone. First, an operations demonstration: the dispatch rule running against a higher-fidelity rover simulator with terrain, lighting and a lander manifest for a real Phase-2 mission, which is the TRL 6 artefact NASA's Appendix B asks for. We estimate this at one engineering year and well under \$1 million, most of it integration with an existing simulator such as the CARLA-based lunar environment used for NASA's autonomy challenge. Second, a payments and gain-sharing layer: the auction here pays variable cost and reports revenue shares; a real market needs prices that owners will accept and a rule for splitting pooled gains that keeps every owner better off than in-house, for which the collaborative-routing literature offers Shapley-value and proportional rules [9, 10]. Third, strategic behaviour: our vehicles bid their true cost. Whether owners would, and what the mechanism should do if they do not, is an open question that matters more as the market thickens.

8. Limitations

The model is a simplification in ways that could change the numbers, though we think not the ordering of the mechanisms. *Terrain* is a nine-node graph with three slope classes and a fixed tortuosity; real traverse times at a specific site will differ, and we did not model dust, obstacle fields or trafficability by mass, which NASA says make heavy transport "exponentially more difficult" [2]. *Speeds* rest on a 6 km/h autonomous average that is our assumption and a teleoperation fit to four published data points. *Demand* is Poisson, so it reproduces NASA's annual totals but not the fact that CLPS task orders have slipped by 14 months on average [38]; a real year may have half or twice the landings we draw. *Bulk flows* are our assumption anchored to the IPEX excavation rate; they drive utilisation but not the urgent-cargo results. *Costs* are built from two contract values and a published Starship price; the hauler's cost is a guess, and Section 6.10 shows the cost results scale with it. *Offloading* is a delay, not a resource; cranes, interfaces and lander geometry could create queues we do not see. *Night* is a single block per cycle at a fixed fraction, where real polar sites have several shorter eclipses. *Energy* never bound in these scenarios and may bind for longer traverses. *Vehicles' other duties* are random blocks; real crew schedules are known in advance and a dispatcher could plan around them. The *mechanism* assumes truthful cost bids and does not model payments, safety-zone deconfliction or communication outages. Finally, every result is for one Earth year from an empty base with 20–30 replications; confidence intervals are wide where landings are few, and several marketplace-versus-greedy differences in the thin Phase-1 setting are not distinguishable from zero. We report them that way.

9. Broader impact and reproducibility

A dispatch marketplace decides whose cargo moves first and whose vehicle earns. In our runs the generalised-cost bid favours cheap, slow vehicles for deferrable lots and fast ones for urgent items, which is efficient but means that an owner of small, slow rovers earns only low-value work; a real platform will need a visible gain-sharing rule and an appeal path, or small suppliers will not join. Night-aware dispatch protects vehicles at the cost of delaying cargo; who bears that cost should be explicit in contracts. Operating other companies' vehicles raises export-control questions that we flag but do not resolve. On the positive side, pooling spreads revenue more evenly than bilateral contracting (Section 6.1) and keeps heavy items moving for owners who have no hauler of their own.

Everything needed to reproduce this paper is in the repository: the simulator, the scenario definitions, the seed lists, the experiment runner that produced every CSV, the figure scripts that produced every figure and table, and the parameter notes with sources. Each cell's seeds are 0–29 (0–19 for the latency suite; 0–7 for the weight search); the demand stream uses the cell seed and the operations stream the seed plus 100,003. Runs are deterministic given a seed. A full rerun takes about 15 minutes on a 12-core desktop.

10. Conclusion

We asked whether a lunar logistics marketplace is a real thing or a sentence in a planning document, and built a testbed to find out. Three answers hold across 7,000 simulated years. Pooling vehicles across owners roughly halves the delay of urgent cargo and keeps every owner's heavy items moving, but only when the fleet is fragmented: with one owner the gain is zero, with six it is 23–54 hours on the 90th percentile. The same service level needs four vehicles under the marketplace or batched assignment and six under greedy pooling, which is a \$90 million-a-year difference in a market where the fleet is 99% of the cost; a surface move costs \$300 per kilogram moved or about \$5,000 per kilogram of landed cargo, a few percent of what landing it costs. And three limits are beyond any dispatch rule: 54–63% of landed mass is immovable by LTV-class rovers alone, of which team lifts recover about half and one shared hauler most of the rest; teleoperation from Earth works at lunar latency with a staffed operator pool and collapses at Mars-like latency; and nights, not faults, are the environmental hazard that dispatch must plan around. The next steps are an operations demonstration against a high-fidelity rover simulator, a payments and gain-sharing layer, and a treatment of strategic bidding.

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Appendix A. Parameters, sources and confidence

Confidence: **high** = read in the primary document; **med** = reputable secondary report or primary seen only in summary; **low** = derived or single-source; **assumed** = our choice, swept in Section 6.10 where it matters.

Parameter	Value used	Basis	Confidence
Phase landed mass	~4,000 / ~60,000 / ~150,000 kg; 21 / 24 / 28 landings	Moon Base guide p.3 [1]	high
Recurring logistics	2,000–6,000 kg per crewed mission; 2,500–10,000 kg/yr	[2, 3]	high
Habitation-class elements	12,000–15,000 kg	[2, 3]	high
Pad standoff; farthest site	>1,000 m; up to 5,000 m	[2]	high
Slopes	up to 20°; >10° common	[2]	high
Tortuosity factor	1.3	our choice	assumed
LTV cargo capacity	800 kg full / 1,600 kg reduced performance; 1,500 kg used	[2, 5]; Astrolab 1,500–1,600 kg [55]	high / med
LTV speed	10 km/h design target; 6 km/h autonomous average used	[1]; Astrolab CLV-1 >6 mph, Pegasus >9 mph [4]	med / assumed
LTV cost	\$219–220M task order + ~\$234M MK1 delivery = \$450M	[4]	high / derived
Hauler	10,000 kg, 4 km/h, \$600M, night-capable	“high capacity mobility” gap [1, 5]; Starship \$100k/kg [46]	assumed
Small rover	300 kg, 3 km/h, \$60M	Lunar Outpost HL-MAPP 250 kg / 200 kg / 3 m/s	med / assumed
Lander payloads	CLPS 100–500 kg; MK1 2,000–3,000 kg; large 12,000–30,000 kg	CLPS task orders 0.07–0.475 t [5]; MK1 3 t; MK2 20–30 t	high / med
Offload delay	6–24 h (CLPS), 12–36 h (MK1), 24–72 h (large)	our choice	assumed
Dark share of cycle	0.15 default (about 106 h)	Shackleton rim 90–94% lit [39]; 120 h survival target [1]	med
Night speed factor (night-capable)	0.6	our choice	assumed
Light-time RTT; processing	2.6 s; +2.4 s default	[28, 45]	high / med
Teleop speed model	3.24/(1+0.105·RTT) km/h	fit to 3.24 / 2.56 / 2.03 / 1.76 km/h at 0 / 4 / 6 / 8 s [29]	med (numbers from summary)
Teleop handling factor	1 + 0.25·RTT	139% at 1.5 s, 174% at 3 s [30]	med
Fan-out	supervised 4; autonomous 20	Olsen & Wood 1–9 [32]; mining up to 30 [33]	med
MTBF / MTTR	500/24 h (LTV), 400/36 h (hauler), 300/12 h (small)	chosen to give 92–96% availability as in autonomous haulage [40]; no rover MTBF published [53]	assumed
Other-duty share	0.3 in 48 h blocks	LTVs serve NASA first, hired out between missions [6]; a trade estimate of 25% commercial use	low
Energy	350 / 900 / 120 Wh/km by class	LRV-derived ~95 Wh/km at 700 kg [41], scaled	low

Operator cost	\$1,000 per console hour	our choice; swept \$200–\$5,000	assumed
Value of time by priority	\$5,000 / \$1,000 / \$100 per hour	our choice	assumed
Wear share ω	0.5	our choice	assumed
Return flow	30% of items; 20–60% of mass; 5–30 days later	trash and sample return are Appendix R topics [7]	assumed
Bulk flows	P1 30 t/yr; P2 500 t/yr; P3 1,500 t/yr	IPEX 10 t/day [36]; swept 0–3,000	assumed
Landed cost per kg	CLPS ~\$1.08M (Blue Ghost M1); MK1 ~\$78k if full; Starship \$100k published	[47]; [4]; [46]	high / derived / med

Table A2. Mission manifest behind Figure 3A (base case, tonnes of surface payload at or near the south pole). Non-polar CLPS flights are excluded. Dates reflect NASA’s May–June 2026 statements and the New Glenn loss of 28 May 2026.

Year	Low	Base	High	Base-case content
2026	0.0	0.5	0.7	Griffin-1 with the FLIP rover (~0.5 t)
2027	0.2	0.8	3.5	Blue Moon MK1 Pathfinder, IM-4, Draper CP-12, VIPER on a second MK1
2028	1.0	4.5	12	Two LTV deliveries on MK1 with co-manifested cargo; Artemis IV crew logistics; MoonFall drones
2029	1.5	6	15	Recurring logistics begins (NASA: 2,500–10,000 kg/yr); Eagle rover on Starship; IM-5
2030	3	20	35	Fission Surface Power unit (≤ 15 t) plus recurring logistics
2031	3	12	30	Recurring logistics; ESA Argonaut (1.5 t); ISRU pilot elements
2032	5	25	45	JAXA pressurised rover (12–15 t class); recurring logistics up to 8 t per mission

Appendix B. STRESS-DES reporting checklist

Item	Where answered
1.1 Purpose of the model	Section 3.1
1.2 Model outputs	Section 3.6, Section 5 (metrics)
1.3 Experimentation aims	Section 1 (RQ1–RQ4), Table 4
2.1 Base model overview diagram	Figure 2
2.2 Base model logic	Sections 3.2–3.6
2.3 Scenario logic	Table 4; experiments/run_all.py
2.4 Algorithms	Section 4, Algorithm 1; lunarsim/dispatch.py
2.5 Components (entities, activities, resources, queues)	Section 3.1, 3.6
3.1 Data sources	Appendix A, References
3.2 Pre-processing	Section 3.4 (latency fit)
3.3 Input parameters	Tables 2, 3, A1
3.4 Assumptions	Appendix A (“assumed” rows), Section 8
4.1 Initialisation and warm-up	Section 5 (empty base, no warm-up deletion; justification given)
4.2 Run length	One Earth year (8,760 h), event-driven
4.3 Estimation approach	30 (or 20) replications; 95% CIs; common random numbers; paired differences
5.1 Software	Python 3.12.3, NumPy, SciPy, NetworkX, pandas, Matplotlib; Windows 11
5.2 Random sampling	NumPy PCG64; demand stream seeded with the cell seed, operations stream with seed + 100,003
5.3 Model execution	Event heap ordered by (time, sequence number); batched mechanisms act at ticks only
5.4 System specification	12-core desktop; 0.1–3 s per simulated year; ~15 min for the suite
6.1 Code access	lunar-logistics-research/ (simulator, scenarios, seeds, figure scripts)

Appendix C. Implementation notes on the mechanisms

Feasibility: a vehicle can carry an item if the item's interface standard is in the vehicle's accepted set and the item's mass does not exceed the vehicle's capacity. Teams: for an item heavier than every free vehicle, candidate vehicles are those that are free, team-capable and compatible, sorted by capacity; the first pair or triple whose capacities sum to at least the item's mass is chosen. Routing within a bundle: all permutations of up to four distinct destinations are evaluated on slope-weighted distance from the pad. Insertion test: an item joins a bundle if its marginal duration is below 80% of the duration of a separate single-item job for the same vehicle. Pool cap: when more than 40 items are waiting, one auction call considers the 40 most urgent and oldest; remaining items are considered at the next event. Payments: each job is paid its variable cost (Section 3.5) and credited to the vehicle owners in equal shares for teams. Batched assignment uses SciPy's `linear_sum_assignment` with $\text{cost} = \text{single-item job duration} - 0.05 \times \text{waiting age}$; infeasible pairs cost 10^6 . In-house carrier assignment: owners without vehicles are mapped round-robin to carriers in sorted order at the start of the run.